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```
sit@sit-Vostro-3400:~/cclab/cn lab$ gcc practical2.c
sit@sit-Vostro-3400:~/cclab/cn lab$ ./a.out
TCP Echo Server listening on port 11000...
Received: hii
Received: how are you
Received: can i talk to you pleaase
```

```
%{
#include <stdio.h>
int lines = 0;
int tokens = 0;
int identifiers = 0;
int keywords = 0;
int whitespace = 0;
%}


KEYWORD    int|float|char|double|if|else|while|for|return|break|continue|void|switch|case|default|do|struct|long|short|unsigned|signed|const|static

ID         [a-zA-Z_][a-zA-Z0-9_]*

%%

{KEYWORD}      { keywords++; tokens++; }

{ID}           { identifiers++; tokens++; }

[0-9]+         { tokens++; }

\"([^\\\"]|\\.)*\"   { tokens++; }

[ \t]+         { whitespace += yyleng; }

\n             { lines++; whitespace++; }

.              { tokens++; }

%%

int yywrap()
{
    return 1;
}

int main()
{
    printf("Enter statement :\n");

    yylex();

    printf("\ncounts:\n");
    printf("Number of lines       : %d\n", lines);
    printf("Number of tokens      : %d\n", tokens);
    printf("Number of identifiers : %d\n", identifiers);
    printf("Number of keywords    : %d\n", keywords);
    printf("Whitespace characters : %d\n", whitespace);

    return 0;
}
```

```
sit@sit-Vostro-3400:~/cclab$ flex practical2.l
sit@sit-Vostro-3400:~/cclab$ gcc lex.yy.c
sit@sit-Vostro-3400:~/cclab$ ./a.out
Enter statement :
my name is aditi rajak. i study in fivth semester of b.tech course.  in btech my stream is computer science. %hello world% printf(hello world)

counts-
Number of lines      : 1
Number of tokens     : 34
Number of identifiers : 26
Number of keywords   : 0
Whitespace characters : 26
sit@sit-Vostro-3400:~/cclab$
```

practical2.l × practical2.c

```c
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>
#include <netinet/in.h>

#define PORT 11000
#define BUF_SIZE 1024

int main() {
    int server_fd, new_socket;
    struct sockaddr_in address;
    char buffer[BUF_SIZE];
    int addrlen = sizeof(address);

    // Create TCP socket
    server_fd = socket(AF_INET, SOCK_STREAM, 0);

    // Prepare server address
    address.sin_family = AF_INET;
    address.sin_addr.s_addr = INADDR_ANY;
    address.sin_port = htons(PORT);

    // Bind
    bind(server_fd, (struct sockaddr *)&address, sizeof(address));

    // Listen
    listen(server_fd, 3);
    printf("TCP Echo Server listening on port %d...\n", PORT);

    // Accept
    new_socket = accept(server_fd, (struct sockaddr *)&address,
                        (socklen_t *)&addrlen);

    while (1) {
        memset(buffer, 0, BUF_SIZE);

        int bytes = read(new_socket, buffer, BUF_SIZE);

        if (bytes <= 0)
            break;

        printf("Received: %s", buffer);

        send(new_socket, buffer, bytes, 0); // echo back
    }

    close(new_socket);
    close(server_fd);

    return 0;
}
```